NETFLOW

LIQUID FLOW MEASUREMENT · NT-UFW SERIES

# WALL-MOUNT ULTRASONIC FLOW METER

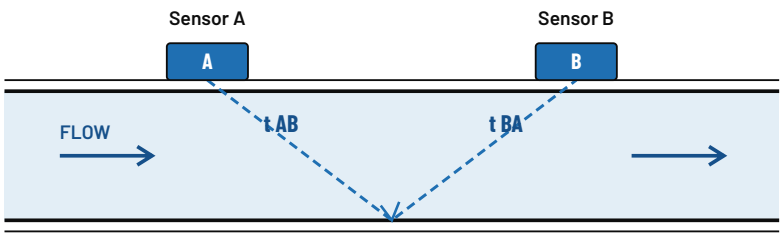
Liquid flow and thermal energy measurement without cutting the pipe or creating pressure loss

The NT-UFW series is a wall-mount ultrasonic flow meter designed for long-term continuous (online) flow measurement. It operates with clamp-on sensors, insertion sensors or in-line pipe-type sensors. It is used for a wide range of liquids—from clean and drinking water to chemicals, from raw wastewater to cooling and river water.

| | |
|---|---|
| **±%1**<br>MEASUREMENT ACCURACY | **DN15 – DN6000**<br>PIPE DIAMETER |
| **IP67 / IP68**<br>CONVERTER / SENSOR | **Pt100**<br>HEAT / ENERGY MEASUREMENT |




## WORKING PRINCIPLE TRANSIT-TIME METHOD




V method · signal reflects from pipe wall

### 1 · SIGNAL TRANSMISSION

Two sensors alternately send and receive ultrasonic signals.

### 2 · TIME DIFFERENCE

The signal in the flow direction (t AB) arrives faster than the one against the flow (t BA).

### 3 · FLOW CALCULATION

The time difference is proportional to the flow velocity; the converter calculates flow from velocity and pipe cross-section.

## FEATURES

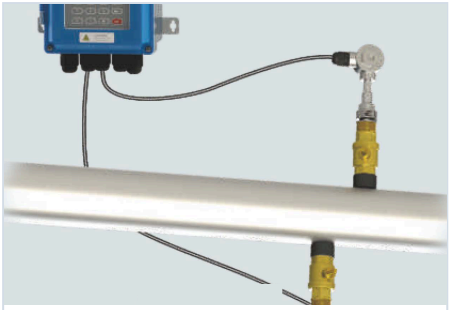
- Non-invasive measurement: no pipe cutting, no water interruption, no pressure loss
- Heat / energy measurement by adding temperature sensor
- No moving parts, long-term stable operation
- Wall, DIN rail or ex-proof enclosure mounting
- Clamp-on, insertion and pipe-type sensor options
- Long-term data logging with optional SD card

## INSTALLATION TYPES



### CLAMP-ON TYPE

Sensors mounted externally. No pipe cutting, no pressure loss; easy and fast installation.



### INSERTION TYPE

Sensors inserted through a valve. Installation without water interruption, stable measurement.



### IN-LINE TYPE

Measuring body installed in-line. Pipe cutting required; highest accuracy and stability.

## SENSOR OPTIONS

SELECTED ACCORDING TO PIPE CONDITION AND INSTALLATION METHOD

| TYPE | MODEL | PIPE DIAMETER | SICAKLIK | DIMENSION |
|---|---|---|---|---|
| Clamp-on | **TS-2 · small** | DN25 – DN100 | –30 ... 90 °C | 45 × 25 × 32 mm |
| | **TM-1 · medium** | DN50 – DN700 | –30 ... 90 °C | 64 × 39 × 44 mm |
| | **TL-1 · large** | DN300 – DN6000 | –30 ... 90 °C | 97 × 54 × 53 mm |
| High temperature clamp-on | **TS-2-HT · TM-1-HT · TL-1-HT** | DN25 – DN6000 | –30 ... 160 °C | Same as standard clamp-on |
| Immersion | **TC-1 standard · TC-2 long** | DN80 – DN6000 | –30 ... 160 °C | 190 / 335 × 80 × 55 mm |
| Boru tipi | **G1 threaded · G2 flanged** | DN15 – DN32 · DN40 – DN1000 | –30 ... 160 °C | SUS304 / carbon steel |

## TECHNICAL SPECIFICATIONS

| | |
|---|---|
| Measuring method | Transit-time |
| Accuracy | ±%1 |
| Pipe diameter | DN15 – DN6000 |
| Protection class | Converter IP67 · sensor IP68 |
| Converter mounting | Duvar · DIN ray · ex-proof kutu |
| Sensor temperature | –30 ... 90 °C · HT modeller –30 ... 160 °C |
| Heat / energy measurement | Pt100 with temperature sensors |
| Data logging | SD kart (optional) |

## ISI MEASUREMENT Pt100

| | |
|---|---|
| **CT-1** | Clamp-on type · >DN50 · no pipe cutting |
| **TCT-1** | Insertion type · >DN50 · pipe cutting required · 100 °C'de ±0,8 °C |
| **PCT-1** | Pressurized immersion · >DN50 · no pipe cutting |
| **SCT-1** | Small diameter pipe · <DN50 · pipe cutting required |

All temperature sensors operate in the range –40 ... 160 °C.

### APPLICATION AREAS

Drinking and utility water · ultra-pure liquids · chemicals · raw and treated wastewater · cooling water · river water · plant effluents · heating / cooling energy measurement

## HEAT / ENERGY MEASUREMENT

UPPER LINE SUPPLY · LOWER LINE RETURN



### CLAMP-ON + CT-1

Flow and temperature sensors mounted externally. No pipe cutting, no pressure loss.



### DALDIRMA + Pt100

Installation without water interruption. Long-term stable and reliable measurement.



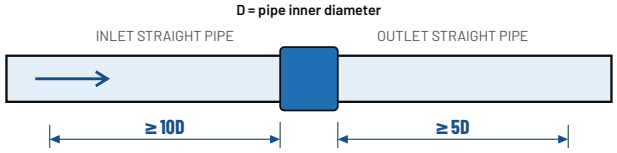
### IN-LINE TYPE + Pt100

Pipe cutting required. Highest accuracy and stability.

The converter evaluates the flow rate together with the temperature difference between supply and return lines to calculate the energy transferred in the heating or cooling circuit.

## INSTALLATION KURALLARI

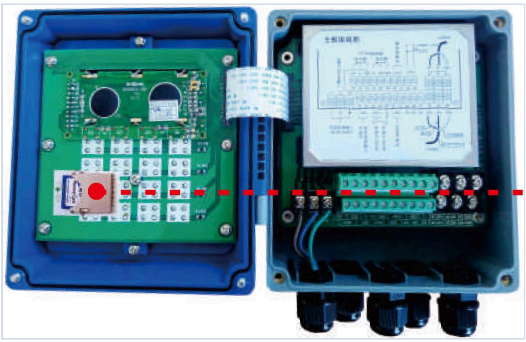
CLAMP-ON AND INSERTION TYPE




A distance of at least 30D must be left after the pump outlet.

- Pipe must be full at measurement point; avoid highest point and free-discharge outlets.
- Prefer vertical lines with bottom-to-top flow.
- On horizontal pipes, place sensors on the side; air on top and sediment below distort measurements.
- For clamp-on type, clean pipe surface and apply conductive gel between sensor and pipe.
- Sensor spacing is set according to the converter's calculation based on pipe diameter and material.
- Mount converter away from frequency inverters and strong electromagnetic sources.

## CONVERTER AND DATA LOGGING



Optional SD card module provides long-term high-volume data logging. Recorded data can be converted to tables, statistics and reports; flow curves can be generated.

- LCD display with keypad
- IP67 converter enclosure
- Front panel parameter setting
- Bottom cable glands

Contact our technical team for sensor selection suitable for pipe diameter, material and fluid.

**+90 541 373 79 53**
**info@netflw.com**